MIL-C-60602B (AR)
16 MARCH 1981
SUPERSEDING
MIL-C-60602A (MU)
3 June 1975

MILITARY SPECIFICATION

CASE IMPELLER, PROJECTILE BURSTER, M175

This specification is approved for use by the US Army Armament Research and Development Command, Department of the Army, and is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the manufacture of the Case Impeller, Projectile Burster, M175.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

☆U.S. GOVERNMENT PRINTING OFFICE: 1981—703-023/1667

FSC: 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

10523973 - Case Impeller, Projectile Burster, M175
10534151 - Plug

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1 Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 General.-Materials and parts shall comply with the requirements specified on the applicable drawings and specifications.

3.2 Surface preparation and application of protective finish.-Where cleaning and surface pretreatment are required, the process shall apply to all indicated surfaces regardless of contour configuration. The protective finish shall be as specified on the applicable drawings and shall be uniformly applied to the surfaces indicated by the applicable drawing

3.3 Soundness of case impeller.-The burster casing shall withstand an internal air pressure of one hundred pounds per square inch (psi) for sixty seconds, minimum (min.), when tested as specified in 4.5.2.

3.4 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship.-The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the following:

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3.5.1 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing and shall be in accordance with the provisions of MIL-A-2550.

3.5.2 Burr.-No part shall have a burr.

3.5.3 Metal defects.-The impeller case and plug shall be free from cracks, splits, inclusions, porosity, seams, laps, folds, workmarks or dents.

3.5.4 Foreign matter.-The impeller case and plug shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be performed on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in the sample quantities indicated:

Part Description	Drawing	Quantity
Case Impeller, Projectile Burster, M175	10523973	25
Plug	10534151	25

4.3.2 Inspections to be performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Case Impeller, Projectile Burster, M175				See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Case Impeller, Projectile Burster, M175 (Dwg. 10523973)				
	Examination for defects	25		3.1	4.4.2.1, 4.4.2.2
	Air pressure test	25		3.3	4.5.2
	Plug (Dwg. 10534151)				
	Examination for defects	25		3.1	4.4.2.3

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plans.-Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. Visual inspections.-In the event of a dispute, four power (4X) magnification shall be used in performing visual inspections for such defects as burrs and foreign matter. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60602B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER
4.4.2.1	Case, Impeller, Projectile Burster		10523973
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Diameter of press fit surface			3.1	Gage
2	Reverse taper on press fit surface			3.1	Gage
3	Press fit radius at front of press fit surface not tangent with press fit surface			3.1	Gage
4	Longitudinal scratches on press fit surface			3.1	Visual
5	Metal defective			3.5.3	Visual
6	Surface finish of press fit surface improper			3.1	Manual/Gage
7	Air pressure test			3.3	4.5.2/Test
Major					
101	Weight, max.		0.40%	3.1	Scale
102	True position of large cavity with press fit surface		0.40%	3.1	Gage
103	True position of fins with press fit surface		0.40%	3.1	Gage
104	Perpendicularity of rear face with press fit surface		0.40%	3.1	Gage
105	Straightness of small cavity		0.40%	3.1	Gage
106	Pitch diameter of thread, max.		0.40%	3.1	Gage
107	Minor diameter of thread, max.		0.40%	3.1	Gage
108	Diameter of large cavity		0.40%	3.1	Gage
109	Diameter of small cavity		0.40%	3.1	Gage
110	Diameter of fin		0.40%	3.1	Gage
111	Depth of large cavity		0.40%	3.1	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60602B (AR)

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
4.4.2.1	CONTINUATION SHEET		10523973
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
112	Depth of small cavity		0.40%	3.1	Gage
113	Length of effective thread		0.40%	3.1	Gage
114	Length of press fit surface, min.		0.40%	3.1	Gage
115	Location of basic diameter, max.		0.40%	3.1	Gage
116	Thickness of wall at hub, min.		0.40%	3.1	Gage
117	Thickness of wall at fin end of large cavity, min.		0.40%	3.1	Gage
118	Thickness of wall at juncture of large cavity and intermediate diameter, min.		0.40%	3.1	Gage
119	Presence of foreign matter		0.40%	3.5.4	Visual
Minor					
201	Angle between fin blades		0.65%	3.1	Gage
202	Diameter of hub, max.		0.65%	3.1	Gage
203	Pitch diameter of thread, min.		0.65%	3.1	Gage
204	Angle at taper, blending intermediate diameter with hub, min. (if applicable)		0.65%	3.1	Gage
205	Depth of center, max. (if applicable)		0.65%	3.1	Gage
206	Depth of drill point, max. (if applicable)		0.65%	3.1	Gage
207	Length of thread, max.		0.65%	3.1	Gage
208	Length, total		0.65%	3.1	Gage
209	Length of fin, min.		0.65%	3.1	Gage
210	Thickness of fin blades		0.65%	3.1	Gage
211	Radius blending intermediate diameter with hub, max. (if applicable)		0.65%	3.1	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60602B (AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.4.2.1	CONTINUATION SHEET				10523973 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
212	Diameter of thread chamfer		0.65%	3.1	Gage
213	Radii or chamfers missing or incorrect		0.65%	3.1	Visual/Gage
214	Presence of a burr		0.65%	3.5.2	Visual/Manual
215	Surface preparation improper		0.65%	3.2	Test/4.5.1
216	Marking missing, incorrect or illegible		0.65%	3.1	Visual
217	Surface finish improper		0.65%	3.1	Gage
218	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60602B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Case, Impeller, Projectile Burster (Prior to Packing)		10523973
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Air pressure test acceptance mark missing			3.1	Visual
Major	None defined				
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60602B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Plug				10534151
					NEXT HIGHER ASSEMBLY
					10523973
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of thread, min.		0.40%	3.1	Gage
102	Major diameter of thread, min.		0.40%	3.1	Gage
103	Thickness		0.40%	3.1	Gage
104	Thread damaged or not full		0.40%	3.1	Visual
105	Metal defective		0.40%	3.5.3	Visual
Minor					
201	Pitch diameter of thread, max.		0.65%	3.1	Gage
202	Inside hexagon missing		0.65%	3.1	Visual
203	Chamfer missing or incorrect		0.65%	3.1	Visual/Gage
204	Diameter of counterbore, max. (if applicable)		0.65%	3.1	Gage
205	Surface finish improper		0.65%	3.1	Gage
206	Presence of a burr		0.65%	3.5.2	Visual/Manual
207	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60602B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Carton, Packed, Prior to Sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Any support missing or damaged		0.65%	3.1	Visual
202	Separator(s) missing or damaged		0.65%	3.1	Visual
203	Improperly packed		0.65%	3.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60602B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Carton, Packed, After Sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Contents loose		0.65%	3.1	Manual
202	Carton damaged, cut or gouged through all layers		0.65%	3.1	Visual
203	Carton not properly sealed		0.65%	3.1	Visual
204	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Surface preparation and application of protective coating.-The sample size and acceptance criteria for the inspection of surface preparation and application of protective coating shall be in accordance with the applicable specifications. Failure to comply with the applicable specification and drawing requirements shall be cause for taking the corrective action recommended therein (see 4.5.1).

4.4.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see 6.2 and Section 6 of MIL-A-48078).

4.5 Test procedures

4.5.1 Surface preparation and application of protective coating.-The procedures for testing the cleaning and surface pretreatment methods for components shall be in accordance with the methods and acceptance criteria for testing as specified in the applicable specification.

4.5.2 Air pressure test.-Each case impeller shall be mounted in an approved test fixture in such a manner that the impeller is completely submerged in water. After the water has been allowed to settle and it has been ascertained that the minimum pressure specified on dwg. 10523973 has been reached, the time requirement for the pressurization without leakage shall be allowed to acrue. Leakage is defined as air bubbles on the surface of the case impeller, as well as, air bubbles which are rising to the surface of the water for the entire time for which the case impeller is required to be pressurized. The contractor shall develop a written operating procedure for this test, including calibration, and shall submit it to the Contracting Officer prior to submission of the first article sample.

4.5.2.1 Calibration.-The air pressure test equipment shall be calibrated prior to the start of each shift of operation and at intervals not exceeding four hours. It shall also be recalibrated prior to re-starting the operation after a stoppage of two hours or more, or following the completion of unscheduled repairs.

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5. PACKAGING

5.1 Packing and marking.-The case impeller and plug shall be packaged, packed and marked, Level B, in accordance with the requirements of MIL-STD-1169.

6. NOTES

6.1 Ordering data.-See MIL-A-48078.

6.2 Submission of designs for approval.-Submit equipment designs as required to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.3 Delivery of data cards to the technical activity.-The following address is to be added to those listed in the contract (Form DD 1423) for delivery of one copy of each Ammunition Data Card prepared by the contractor for the duration of the contract: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-E, Dover, New Jersey 07801.

6.4 Changes from previous issue.-Due to the extensiveness of the changes, asterisks are not used in this revision to identify changes with respect to the previous issue.

Custodian:
ARMY-AR

Preparing Activity:
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Project Number: 1315-A232

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-60602B CASE, IMPELLER, PROJECTILE BURSTER, M175

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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